

Fig. 1: A typical ADSL set-up

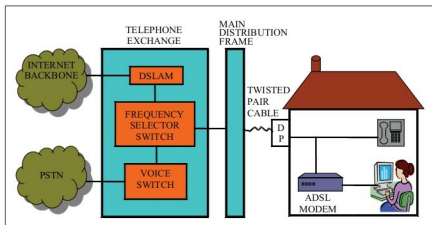


Fig. 2: A typical ADSL Lite set-up

ing download speeds up to 50Mbps and uses the frequency band up to 3.75MHz.

ADSL set-up

Basically, ADSL caters to connections between Internet service providers (ISPs) and subscribers. In order to provide ADSL connections, the existing telephone network is utilised and a network element called digital subscriber line access multiplexer (DSLAM) is installed at the central office/telephone exchange along with a modem placed at the subscriber's home or workplace. The DSLAM works like a concentrator. It provides multiple DSL connections for access to the bandwidth available to the DSLAM from the Internet backbone. The connection to the subscriber is then given from DSLAM via copper lines.

There are two types of DSLAMs—remote DSLAM and CO DSLAM. The remote DSLAM sits in the digital loop carrier (DLC) system in the neighbourhood of the subscriber. On the other hand, CO DSLAM sits in the telco's central office and has a large capacity. The ADSL modem performs the task of converting the ADSL physical-layer signal into a format that can be understood by a computer or any other equipment connected to it. Fig. 1 shows a typical ADSL set-up.

The frequency selector switch at the telephone exchange/central office is essentially a filter that filters out the incoming ADSL signal and diverts the voice traffic to the voice switch. The voice switch is further connected to the public switched telephone network (PSTN). The data traffic is directed towards the DSLAM, which is

connected to the Internet backbone.

The pair from DSLAM is terminated into the main distribution frame (MDF), from where connections are provided to subscribers via twisted copper-pair lines. These twisted pair lines terminate at a DP box placed near the subscriber's home or workplace. The connection from the DP box is first run to the splitter that sits in the customer's premises. The splitter, which is basically a low-pass filter, removes the plain old telephone system (POTS) signal (voice signal, 300Hz-3400Hz) from the incoming ADSL signal. The high-frequency filtered ADSL signal is then given to an ADSL modem. Finally, the connection is taken out from the Ethernet port of ADSL modem and is terminated to the subscriber's computer.

Another variant of ADSL that does not necessitate the need of a POTS splitter is known as splitter-less ADSL and is defined in G.992.2 or G.Lite. ADSL Lite provides downstream speeds of 1.5Mbps and upstream speeds of 512kbps. Reduced data rates cause less interference in POTS signal, which provides freedom from a splitter at subscriber's premises. So, it is possible to access the Internet by connecting G.Lite modems directly to phone jacks. A typical ADSL Lite set-up is shown in Fig. 2. This lower-version of ADSL results in significant cost-saving that can be passed on to subscribers. The plug-and-play functionality makes ADSL Lite more attractive to subscribers.

ADSL working principle

ADSL communication is a full-duplex communication system that is achieved on a pair of wires either by frequency division duplex (FDD), time division duplex (TDD) or echo-cancelling duplex (ECD). In FDD, two separate frequency bands are used, namely, upstream and downstream frequency bands. The modulation technique used in ADSL is either orthogonal frequency division multiplexing (OFDM) or discrete multi-tone (DMT). OFDM and DMT are almost